s9b_inverse_vol_matched

equity strategy | family: equity-volatility | status: validated | derived from `s9b_inverse_vol`

What this strategy does

9b with base unit solved so annualized vol equals Strategy 6's (4.94%)

Why it should work. PEAD is a stronger, cleaner signal in low-vol stocks; size positions by inverse risk.

Design notes (from the strategy's source file).

Strategy 9b -- Strategy 6 sized by inverse idiosyncratic volatility.

Within every (quarter, side, sector, size) cell, each event's weight is scaled by 1/ivol, so calm stocks get more capital than wild ones. Rationale: the earnings drift is worth more per unit of risk in low-vol names (the SUE x vol interaction is t = -2.4 to -2.7 in Fama-MacBeth), and the engine's cost/liquidity model charges the same bps regardless of vol. Raises Sharpe 1.30 -> 1.47 with no fitted parameter, in all three sub-eras.

`InverseVolVolMatched` re-sizes it to Strategy 6's exact volatility to answer "at the SAME risk, how much more return": 7.31% vs 6.49%.

parameter	value
base_unit_fraction	0.003468808347716693
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

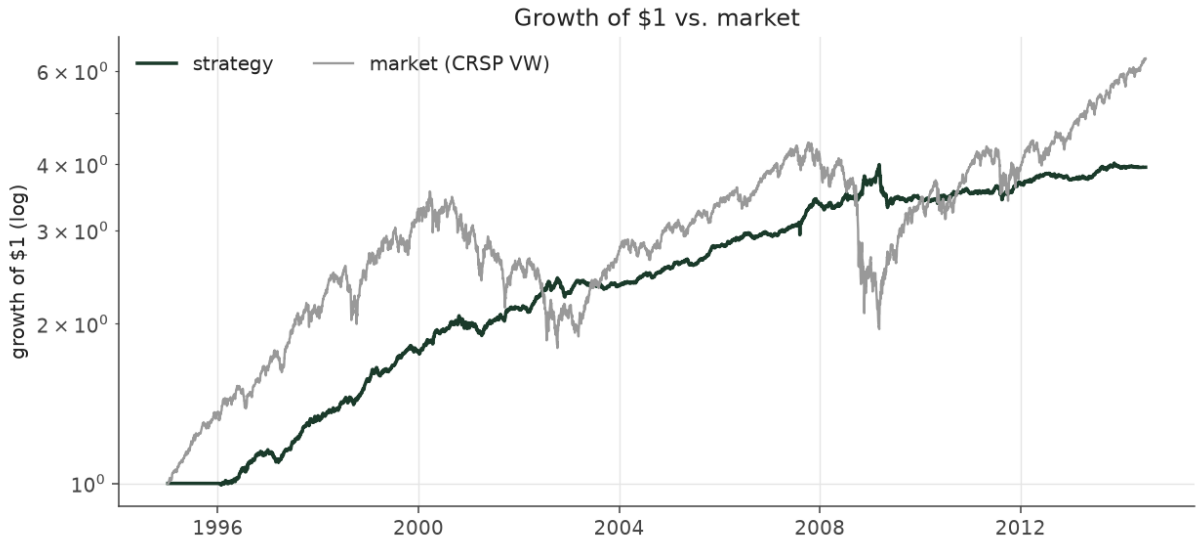
metric	s9b inverse vol matched	market (CRSP VW)
Annualized return	7.31%	9.94%
Annualized volatility	4.94%	19.35%
Sharpe (NAV-pct)	1.45	0.59
Sortino	2.15	
Max drawdown	-17.0%	-55.5%
Calmar	0.43	
Final NAV ($10M start)	$39,500,315	
Worst day	-3.35%	
% positive days	52.4%	
Sharpe 1996-2001	2.39	
Sharpe 2002-2007	1.66	
Sharpe 2008-2013	0.52	

Correlation of daily returns with the market: **-0.00**.

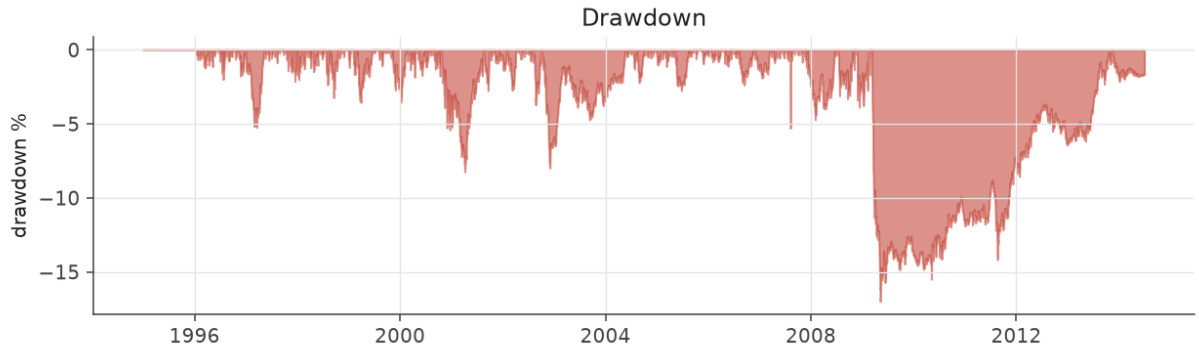
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.45

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0731

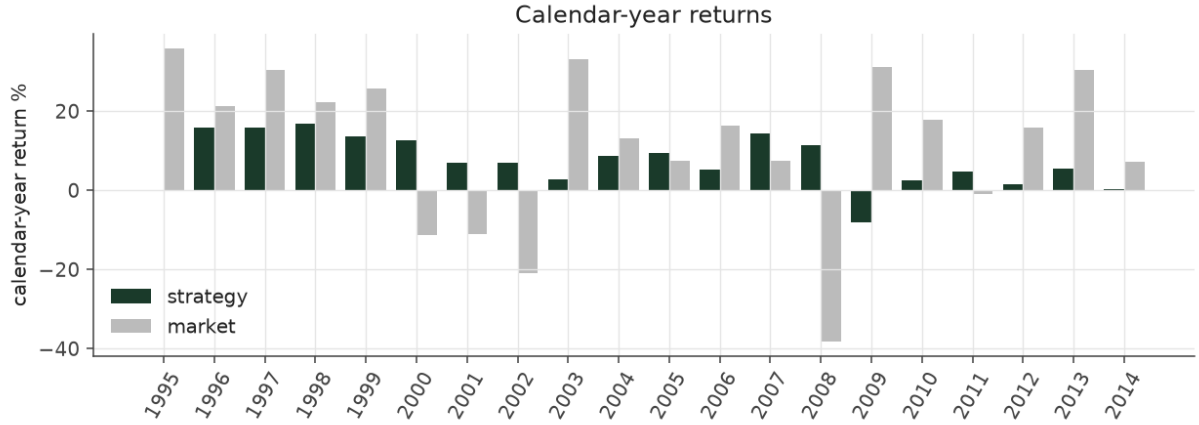
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.43



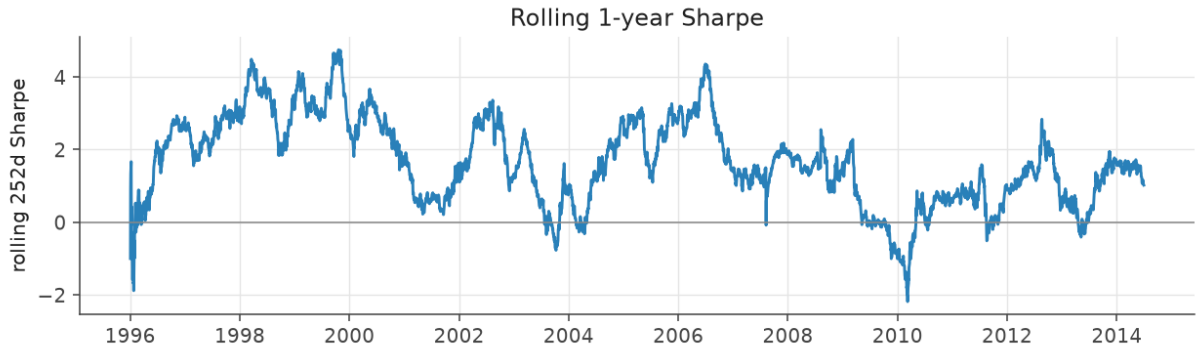
NAV (log scale)



Drawdown from running peak

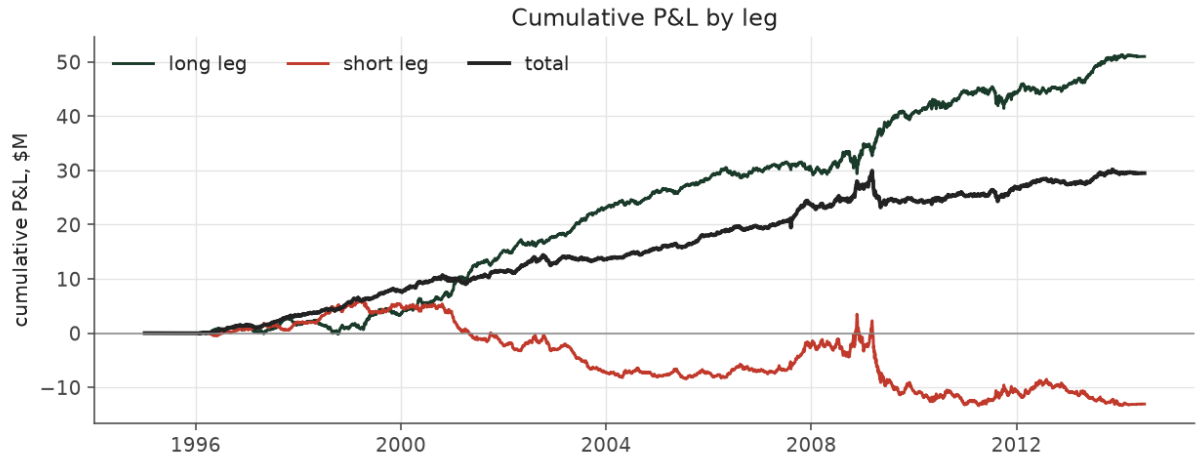


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	51.01	-13.08
P&L after costs ($M)	46.96	-17.46
of which common (average-event) component ($M)	33.69	-34.49
of which excess / PEAD component ($M)	17.32	21.41
excess minus costs ($M)	13.27	17.03
gross return per $ traded (bp)	242.65	-60.53
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.32	-0.23

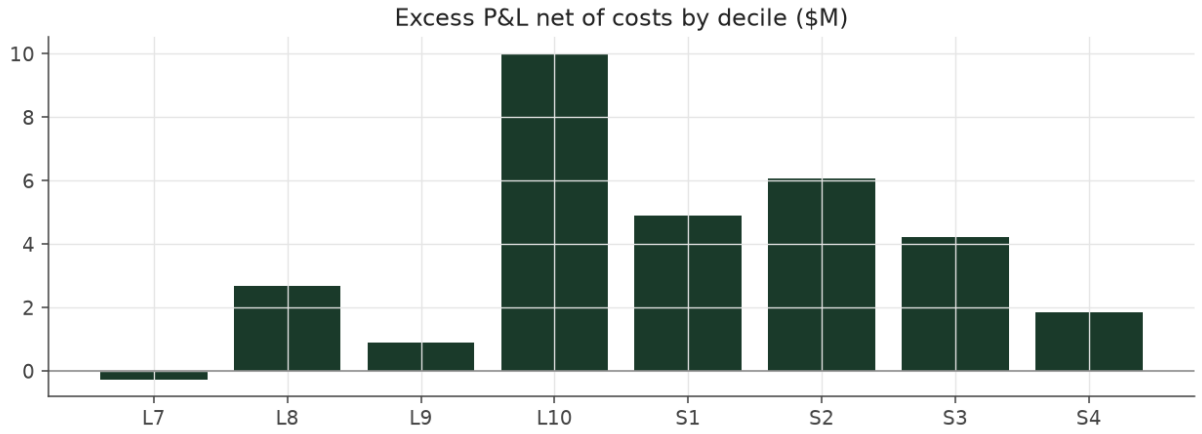
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-13.1M before costs) while the long leg earns +51.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 33.7M / 34.5M. Stripping that common component, the surprise signal itself earned +17.3M on the long side and +21.4M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.32 long / -0.23 short vs 1.45 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,017	187.53	3.24	0.08	0.36	-0.28	-0.9%	-15.12
long	8	16,581	496.58	10.87	3.63	0.96	2.67	8.8%	53.76
long	9	17,745	651.92	10.47	2.12	1.24	0.88	2.9%	13.48
long	10	17,173	766.04	22.39	11.49	1.48	10.01	33.0%	130.65
short	1	17,166	766.10	-6.38	6.51	1.61	4.90	16.2%	63.93
short	2	17,276	669.92	-5.27	7.41	1.34	6.07	20.0%	90.63
short	3	15,967	514.75	-4.22	5.23	1.02	4.21	13.9%	81.69
short	4	9,161	209.99	-1.58	2.25	0.40	1.85	6.1%	88.25



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,194	334.35	5.62	7.77	1.28	6.49	21.4%	194.22
2	22,880	766.10	6.70	9.33	2.16	7.18	23.7%	93.68
3	25,498	978.87	7.90	9.75	2.06	7.68	25.4%	78.51
4	27,576	1,082.62	5.42	6.98	1.71	5.26	17.4%	48.63
5	28,938	1,100.90	3.87	4.89	1.21	3.68	12.1%	33.44

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,631	247.89	5.24	6.96	1.24	5.72	18.9%	230.70
2	23,084	835.03	6.57	10.23	2.50	7.72	25.5%	92.48
3	25,848	1,073.98	8.16	10.90	2.15	8.75	28.9%	81.51
4	27,653	1,066.69	5.75	6.35	1.49	4.86	16.0%	45.55

5	28,865	1,038.89	3.74	4.24	1.04	3.20	10.6%	30.79

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,837	1,657.83	0.68	9.89	2.91	6.98	23.0%	42.10
41-60d	25,647	973.48	6.02	6.04	1.96	4.09	13.5%	41.97
61-90d	35,313	1,222.87	8.62	10.36	2.55	7.81	25.8%	63.84
>90d	12,289	408.66	14.18	12.44	1.01	11.43	37.7%	279.64

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	809.38	11.18	13.03	1.57	11.46	37.8%	141.54
2002-07	39,913	1,427.49	12.27	14.91	2.77	12.15	40.1%	85.09
2008-13	45,278	2,025.98	6.06	10.79	4.09	6.70	22.1%	33.06

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,085	427.07	2.45	3.33	0.90	2.43	8.0%	56.95
Chems	2,588	279.38	1.84	2.28	0.48	1.80	5.9%	64.39
Durbl	2,843	300.63	2.59	3.17	0.53	2.64	8.7%	87.71
Enrgy	6,194	342.20	4.51	5.29	0.63	4.67	15.4%	136.36
Hlth	11,685	411.81	3.00	3.85	0.86	3.00	9.9%	72.72
Manuf	12,033	383.08	2.04	3.02	0.78	2.24	7.4%	58.46
Money	17,569	375.06	3.08	4.03	0.83	3.20	10.6%	85.34
NoDur	5,080	341.14	1.72	2.34	0.63	1.70	5.6%	49.90
Other	18,372	410.74	3.06	4.07	0.91	3.16	10.4%	76.96
Shops	11,984	398.66	3.75	4.58	0.81	3.77	12.4%	94.55
Telcm	3,981	327.17	0.81	1.44	0.58	0.86	2.8%	26.17
Utils	4,672	265.90	0.66	1.32	0.48	0.84	2.8%	31.68

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,494	1,057.22	4.66	7.73	1.92	5.82	19.2%	55.02
low vol	40,087	1,759.86	13.57	17.56	3.70	13.86	45.7%	78.77
mid vol	37,893	1,388.11	11.06	13.09	2.69	10.40	34.3%	74.93

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.31%	0.00%
0.5%	2.2	7.00%	0.31%
1.0%	4.3	6.67%	0.64%
2.0%	8.7	5.95%	1.36%
5.0%	21.7	3.01%	4.30%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.